K.R. Mangalam University - School of Engineering & Technology

Task 3: ML in Cyber Security - Comparative Case Study

Problem domain: Credit Card Fraud Detection

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1. Introduction and Scope

Credit card fraud is a financial cyber-security problem: stolen or misused card data is used for unauthorised transactions. Machine learning is used to score every transaction as legitimate or fraudulent in real time. This case study compares **three ML techniques applied to the same domain and the same public dataset**: **Logistic Regression** (linear model), **Random Forest** (bagging ensemble) and **XGBoost** (gradient-boosting ensemble). The performance figures below are the ones **reported in the cited papers**; no new experiments were run for this study.

2. Common Dataset: ULB / Worldline Credit Card Fraud Dataset

- Published by the Machine Learning Group of Universite Libre de Bruxelles (ULB) with Worldline (Dal Pozzolo et al., 2015); freely available on Kaggle.
- 284,807 transactions made by European cardholders over two days in September 2013, of which only **492 (0.173%) are fraudulent** - an extreme class imbalance.
- Features: V1-V28 (anonymised by PCA for privacy), Time, Amount and the Class label (0 = legitimate, 1 = fraud).
- In the main reference study the data was split 75:25 with stratification, giving a test set of 71,079 legitimate and 123 fraudulent transactions; models were evaluated on this original, imbalanced test set.

3. Technique-wise Analysis

3.1 Logistic Regression (LR)

Algorithm description. A linear probabilistic classifier. It computes a weighted sum of the input features and passes it through the sigmoid function to obtain the probability that a transaction is fraud; a transaction is flagged when the probability is above a threshold (0.5 by default). Weights are learned by minimising log-loss.

Dataset used. ULB credit card fraud dataset (284,807 transactions, 492 frauds).

Performance reported in literature. Popova & Gardi (2025), imbalanced training data: precision 84.8%, recall 63.4%, F1 72.6% (weakest of the models compared). With hybrid sampling (fraud raised to about 1-2.5% of training data) recall rose to 78.1% and F1 to 76.2% while precision fell to 74.4%. When trained on a 1:1 undersampled set and tested on the original data, precision collapsed to 5.0% (recall 90.2%). A separate study on a PCA-transformed dataset with a 5% fraud ratio reports recall 0.90 but precision only 0.68.

Advantages

- Simple, fast to train and to score - suitable for real-time decisions.
- Highly interpretable: coefficients show how each feature raises or lowers fraud probability.
- Needs little tuning and works as a baseline; recall improves substantially with resampling.

Limitations

- Linear decision boundary cannot capture complex, non-linear fraud patterns.
- Most sensitive to class imbalance; low recall (63.4%) on raw data means about 45 of 123 frauds were missed.
- Training on artificially balanced data gives many false alarms on real data.

3.2 Random Forest (RF)

Algorithm description. An ensemble of many decision trees. Each tree is trained on a bootstrap sample of the data (bagging) and a random subset of features at each split; the final prediction is the vote (or average) of all trees. Averaging reduces the variance and over-fitting of a single tree.

Dataset used. ULB credit card fraud dataset (284,807 transactions, 492 frauds).

Performance reported in literature. Popova & Gardi (2025), imbalanced training data: precision 94.2%, recall 78.9%, F1 85.8% - the best F1 of all five models in that study (97 of 123 frauds detected). Hybrid sampling gave recall 83.7%, F1 86.2%, precision 88.8% (103 frauds detected). A SMOTE-based comparison (International Journal of Safety and Security Engineering, 2025) reports accuracy 0.999, F1 0.872, ROC-AUC 0.978 and AUPRC 0.871 for Random Forest.

Advantages

- Handles non-linear relationships and feature interactions with no scaling needed.
- Robust to noise and over-fitting; strongest results on the imbalanced data in the reviewed studies.
- Provides feature importance and works well with the default parameters.

Limitations

- Larger memory use and slower prediction than a single model (many trees), which matters for real-time scoring at scale.
- Less interpretable than Logistic Regression.
- Still misses about 15-20% of frauds, and its probability scores are not always well calibrated.

3.3 XGBoost (Extreme Gradient Boosting)

Algorithm description. A gradient-boosting ensemble in which trees are built sequentially: each new tree is fitted to the errors (gradients of the loss) of the previous trees. Regularisation (L1/L2 penalties), shrinkage and column sampling control over-fitting, and it is optimised for speed and handles missing values.

Dataset used. ULB credit card fraud dataset (284,807 transactions, 492 frauds).

Performance reported in literature. Popova & Gardi (2025), imbalanced training data: precision 94.0%, recall 76.4%, F1 84.3% (94 of 123 frauds detected); hybrid sampling gave recall 82.1%, F1 85.6%, precision 89.4%. The SMOTE comparison study (2025) reports accuracy 0.999, F1 0.837, ROC-AUC 0.983 and AUPRC 0.867. When trained on fully SMOTE-balanced data and tested on the original data, XGBoost precision dropped to 16.5% (recall 86.2%).

Advantages

- High accuracy and very good recall/F1; the highest ROC-AUC (0.983) in the SMOTE study.
- Efficient, scalable and parallel; built-in regularisation and missing-value handling.
- Class-weighting options (scale_pos_weight) suit imbalanced data.

Limitations

- Many hyper-parameters need careful tuning; results vary with the settings.
- Less transparent than LR; can over-fit noisy labels.
- Precision can fall sharply if trained on heavily oversampled data (16.5% in the study above).

4. Comparative Summary

Aspect	Logistic Regression	Random Forest	XGBoost
Type	Linear, probabilistic	Bagging ensemble of trees	Boosting ensemble of trees
Precision (imbalanced training)	84.8%	94.2%	94.0%
Recall	63.4%	78.9%	76.4%
F1-score	72.6%	85.8%	84.3%
Hybrid sampling: P / R / F1	74.4 / 78.1 / 76.2	88.8 / 83.7 / 86.2	89.4 / 82.1 / 85.6
Interpretability	High	Medium	Medium-low
Training / scoring cost	Lowest	Medium-high	Medium
Handles non-linearity	No	Yes	Yes
Sensitivity to imbalance	Highest	Lowest of the three	Moderate

Source for the first block of rows: Popova & Gardi (2025); fraud class, 75:25 stratified split, tested on the original imbalanced test set (123 frauds).

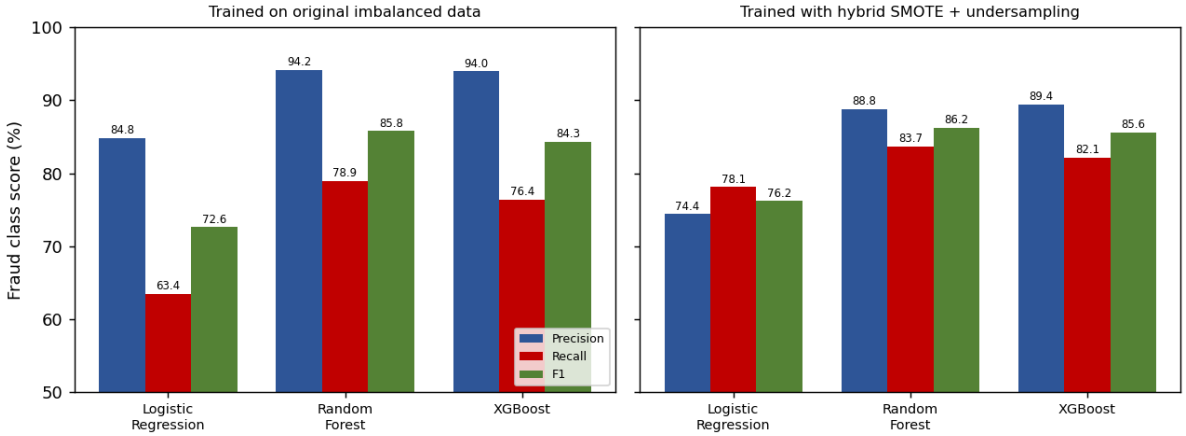


Fig 1: Fraud-class precision, recall and F1 reported by Popova & Gardi (2025). Values redrawn from the paper's tables.

5. Discussion

Ensemble tree methods (Random Forest and XGBoost) clearly outperformed Logistic Regression in the reviewed studies, mainly in recall - the share of real frauds caught - which is the most important metric because a missed fraud is a direct financial loss. Random Forest was slightly better than XGBoost in the main reference study, while XGBoost had a higher ROC-AUC in the SMOTE study, so the ranking of the two ensembles depends on the data split, sampling method and tuning. The reviewed papers also agree on three practical points: (1) **accuracy is misleading** (all models exceed 99.9% accuracy because 99.8% of transactions are legitimate), so precision, recall, F1 and AUPRC must be used; (2) **resampling** improves recall but can reduce precision, and it must be applied to the training set only; (3) models must be evaluated on the **original imbalanced test set** to reflect real deployment, where a flood of false alarms inconveniences customers.

6. Common Limitations of the Reviewed Work

- The dataset covers only two days of 2013 transactions, so it may not generalise to current fraud patterns (concept drift).
- V1-V28 are PCA-anonymised, which prevents business interpretation and feature engineering.
- Only 492 frauds exist, so the test set has just 123 frauds; one more or fewer detected fraud changes recall by about 0.8 percentage points, and figures from different studies (different splits and seeds) cannot be compared exactly.
- The studies are offline experiments; real systems also need latency limits, cost-sensitive thresholds and human review of flagged transactions.

7. Conclusion

For credit card fraud detection, Logistic Regression is a fast and interpretable baseline but misses many frauds on imbalanced data, while Random Forest and XGBoost reach F1 scores of about 84-87% on the same dataset and are the preferred choices. The best system in practice would combine a tuned ensemble with class-imbalance handling, threshold selection based on fraud cost, and analyst review of alerts.

8. References

- I. Popova and H. A. A. Gardi, "Credit Card Fraud Detection," arXiv:2509.15044, 2025. https://arxiv.org/pdf/2509.15044
- "Comparative Analysis of Machine Learning Models for Credit Card Fraud Detection Using SMOTE for Class Imbalance," International Journal of Safety and Security Engineering, vol. 15, no. 5, 2025. doi:10.18280/ijsse.150504
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- Comparative study of Logistic Regression, Random Forest, XGBoost and Soft Voting on a PCA-transformed credit card dataset with 5% fraud ratio (Atlantis Press). https://www.atlantis-press.com/article/126025191.pdf